

Agri-Sovereign / Uzhavan-Sahayak

24-Hour Preliminary Hackathon Execution Plan
Application-first MVP + focused LLM validation

1. Goal for Tomorrow

Win the qualifier with a usable farmer application. Do not make full-scale 8B training the critical path. Build the end-to-end product first; run a small Ministral 3B adaptation experiment in parallel if compute/time permits.

Priority	Deliverable	Done when...
P0	Farmer MVP	Tamil text/voice query produces a useful response.
P0	WhatsApp	Farmer can send a message and receive an answer.
P0	Tamil voice	Voice → ASR → answer → optional TTS.
P0	Agricultural RAG	Answer is grounded in authoritative documents.
P0	Safety validator	Chemical/dosage output is PASS/FAIL/REVIEW.
P1	Ministral 3B	Base model runs on NVIDIA vGPU.
P1	Adaptation	Small QLoRA/CPT pilot completes if feasible.
P1	Evaluation	20–50 questions compared before/after.
P1	Tokenizer	Tamil tokens/word measured; no premature vocabulary expansion.

2. Architecture to Build

Frontend: Next.js + React + TypeScript **Backend:** Python + FastAPI **Model:** Ministral 3B Base **Knowledge:** RAG **Voice:** Tamil ASR/TTS **Messaging:** WhatsApp Business **Safety:** deterministic validator.

Flow: Farmer → WhatsApp/Web → FastAPI → ASR if voice → retrieve agricultural evidence → Ministral → safety validation → structured Tamil answer → text/audio response.

3. Team Checklist

Team	Immediate responsibility
Frontend	Farmer dashboard/chat, voice upload, source and safety cards.
Backend	FastAPI orchestration, sessions, model endpoint, RAG endpoint.
WhatsApp/Voice	Webhook + Tamil ASR + TTS + message/audio handling.
Data/RAG	Inspect 70 GB, clean subset, chunk documents, build retrieval index.
LLM	Ministral setup, baseline, QLoRA/CPT pilot, evaluation.
Safety	Crop/pest/chemical/dosage verification and escalation.
Pitch/Demo	Demo script, screenshots, measured metrics, architecture slide.

4. 24-Hour Checklist

- Get vGPU model, VRAM, GPU count, CUDA, RAM and disk.
- Locate the 70 GB data; never modify the originals.
- Inspect formats, document count, Tamil/English share and estimated tokens.
- Clean: corrupted/empty files, language filtering, deduplication and quality filtering.
- Measure Ministral Tamil token fertility on a representative sample.
- Create 20–50 Tamil agricultural evaluation questions with verified expected facts.
- Run the unmodified Ministral baseline; save answers, latency and token counts.
- Build RAG from the strongest authoritative agricultural material.
- Implement deterministic chemical/dosage safety validation.
- Build Next.js farmer UI and FastAPI backend.
- Connect WhatsApp webhook and Tamil ASR/TTS.
- If GPU/time permits: run a small QLoRA or CPT pilot; do not block the MVP.
- Run identical questions after adaptation and produce before/after metrics.
- Freeze one reliable demo scenario and rehearse it.

5. LLM Experiment — Minimum Defensible Version

Model: Ministral 3B Base. First measure the existing tokenizer. Then run a small domain-adaptation pilot. Use full CPT only if the vGPU allocation genuinely supports it; otherwise use QLoRA as the qualifier prototype. The full 8B + multi-billion-token training remains the scale-up phase.

Stage	Action	Evidence
1	Model smoke test	Model loads and generates on vGPU.
2	Tokenizer test	Measured Tamil tokens/word.
3	Baseline	50 agricultural answers from base model.
4	Adaptation	Small QLoRA/CPT run with loss/checkpoints.
5	After test	Same questions, same scoring.
6	Report	Before/after table; only measured numbers are claimed.

6. What NOT to Do

- Do not attempt full 8B + 4.5B-token CPT in the qualifier.
- Do not spend the whole day preprocessing all 70 GB.
- Do not add 24,000 Tamil tokens before measuring fertility.
- Do not build every multimodal feature if it threatens the core demo.
- Do not present proposal targets as achieved results.

7. Demo Story

One reliable scenario: Farmer sends a Tamil voice message → system transcribes it → retrieves relevant agricultural evidence → Ministral generates a structured answer → safety validator checks any chemical/dosage recommendation → farmer receives Tamil text + optional audio. Show the retrieved evidence and safety status on screen.

8. Jury Defense

- **Why 3B?** It is the practical prototype size for the available qualifier compute; the same adaptation pipeline scales to the 8B target.
- **Why RAG?** It supplies current, authoritative evidence; model adaptation improves domain/language behavior. They are complementary.
- **Why safety validation?** The LLM is not treated as the final authority for safety-sensitive recommendations.
- **What is novel?** Tamil lexical/tokenization analysis + continual domain adaptation + agricultural grounding + safety-validated deployment.
- **What have we proven?** The product workflow plus measured baseline/adaptation evidence—not that the full-scale foundation model is already trained.

9. Definition of Success

Qualifier success: a farmer can use the system end-to-end, the answer is grounded in agricultural evidence, unsafe chemical/dosage outputs are intercepted, and at least one ML result is measured and reproducible.